AI, governance and cost

Microsoft Fabric

C8 Copilot and AI in Fabric

C8-L1 Copilot in Fabric, where it runs and what it costs

Confirm Copilot is enabled on a paid capacity, record the release status of each experience, and budget its CU use.

Copilot in Fabric capacity requirement	Paid F2+ or P1+ in a supported region, with tenant settings enabled. Not after a trial expires.
Copilot release status	Per experience. Notebooks and Warehouse SQL preview. Data agents, Data Factory and Power BI GA (September 2026).
Copilot token rates per 1,000 tokens	Input 100, cached input 10, output 400 CU seconds. Background job, smoothed over 24 hours.
Copilot output	Sometimes inaccurate. A person able to judge it reviews it before use.

C8-L2 Data agents, questions in plain language

Narrow the schema, define your terms, add validated examples, and measure accuracy against known answers.

Data agent query languages	SQL for lakehouse and warehouse, DAX for semantic models, KQL for KQL databases. Up to five sources.
Data agent security	Read-only queries only. Runs with the asking user's permissions.
Example queries	Question and query pairs, validated against the schema. Not supported for semantic models.
Measure a data agent	Ground-truth questions with known answers, rerun after every change, accuracy recorded.

C8-L3 Verify what Copilot writes

Read it, run it on data with a hand-worked answer, compare it with a second method, and keep those checks as tests.

A Copilot query ran without error. What does that prove?	Syntax parses and names exist. Nothing about correctness.
Four checks for Copilot output	Read every line. Run on known data. Compare with a second method. Keep the checks as tests.
Average of averages	Weights small groups like large ones. Divide total by total instead.
Known data for verification	Five to ten hand-built rows with a null, a zero denominator, a duplicate key and an empty group.

C9 Governance and security

C9-L1 Domains and endorsement, finding the right data

Domains help people find data. Endorsement tells them which copy to trust. Neither controls access.

What a domain does	Groups workspaces by business area for discovery and delegated settings. Does not control access.
Domain roles	Fabric admin creates domains. Domain admin edits one. Domain contributor, a workspace admin, assigns their workspaces.
Endorsement badges	Promoted (creators), Certified (named reviewers), Master data (named users, data items only). Dashboards get none.
Does endorsement block use of other copies?	No. It adds a badge and ranks the item first in some lists.

C9-L2 OneLake security, roles down to rows and columns

Readers are Viewers, defaults are removed, related row and column rules share a role, and every engine runs in a mode that enforces them.

OneLake security combines roles how?	Within a role: intersection of table, row and column rules. Across roles: union. Row filters OR together.
Whom does OneLake security restrict?	Viewers and Read-permission users. Admins, Members and Contributors already have data access.
SELECT * with a hidden column	Spark returns allowed columns. SQL analytics endpoint returns an error.
Engines enforcing OneLake RLS and CLS (GA)	Spark, lakehouse, Direct Lake on OneLake, SQL analytics endpoint in user's identity mode. Eventhouse: RLS only, preview.

C9-L3 Sensitivity labels, Purview and lineage

Run impact analysis before changes. Label sensitive items, attach protection policies, and let inheritance carry the label downstream.

Downstream label inheritance rules	Flows to downstream items. Never overwrites a manual label. Never downgrades.
Protection policy	Tied to a label. Named users and groups keep existing permissions. Everyone else is blocked.
Lineage view	Every workspace has one. All items in the workspace plus external sources one step upstream.
Impact analysis	Downstream items and workspaces a change might affect, with a way to notify their contacts.

C9-L4 Audit evidence, Purview scans, label coverage and the audit log

Answer what you hold, how it is protected and who touched it with dated system evidence. Export audit records before they expire, and restrict who sees scanned HR metadata in Purview.

Three audit questions and their sources	What you hold: a Purview Data Map scan. How it is protected: label coverage in the OneLake catalog. Who touched it: the unified audit log.
Audit (Standard) retention	180 days. Audit (Premium) keeps up to a year by policy, longer with an add-on. Export on a schedule for longer periods.
Who sees scanned Fabric metadata in Purview?	Purview permissions decide, not Fabric roles. Restrict the collection that holds HR items.
What a Fabric scan does not do	It does not classify or label Fabric data. It brings in metadata and lineage.

C9-L5 Attribute-based access, rows by business unit

OneLake security row filters are static. For "each manager sees their own unit", build groups from Entra attributes or join an entitlements table in a Warehouse security policy. Audit who can write the attributes and the table.

RBAC and ABAC	RBAC grants by role membership. ABAC decides from attributes such as department or business unit.
Can a OneLake security row filter depend on the current user?	No. OneLake RLS roles don't support dynamic or multitable queries. Use Entra dynamic groups mapped to roles, or Warehouse RLS.
Dynamic row security in a Fabric Warehouse	`CREATE SECURITY POLICY` with a filter predicate: an inline table-valued function that joins an entitlements table on `USER_NAME()`.
Entra dynamic groups for access: limits and risks	Each member needs Entra ID P1. A group change takes about an hour to reach OneLake. Audit who can write the attributes the rule reads.

C10 DevOps and cost

C10-L1 Git integration for Fabric workspaces

One workspace per developer on its own branch, pull requests into main, and a data reload plan, because Git holds definitions, not data.

What Git integration stores for a lakehouse	Metadata only: name, shortcuts, OneLake security roles. Not tables or files.
Commit vs Update	Commit exports workspace changes to Git. Update imports Git changes and overwrites the workspace version.
Who connects a workspace to Git?	Admin only. Members and Contributors commit and update. Viewers see no Git information.
Recommended development process	One workspace per developer on its own branch. Pull requests into main.

C10-L2 Deployment pipelines, dev to test to production

Stage-specific settings belong in deployment rules. Compare before you deploy. Reload data after, because deployments copy items, not data.

Deployment pipeline stages	2 to 10 stages, default Development, Test, Production. One workspace per stage.

What a deployment copies	Item content and metadata. Not data: refresh models and reload tables after.
Deployment rule types	Data source, parameter, default lakehouse. Defined in the target stage on the item.
Deploying over an existing item	Content is overwritten. ID, URL and permissions stay the same.

C10-L3 Capacity, smoothing and throttling

Know which of your jobs are background, run them off-peak, protect production with surge protection, and read the Metrics app before you resize.

Smoothing windows	Interactive: 5 to 64 minutes. Background: 24 hours. Timepoints are 30 seconds, 2,880 a day.
Throttling stages	Up to 10 min: none. 10 to 60: 20 s interactive delay. 60 min to 24 h: interactive rejection. Over 24 h: all rejected.
Surge protection	Capacity level: earlier background rejection. Workspace level: cap a workspace's CU share over 24 hours, except Mission Critical.
Capacity Metrics app	Installed by a capacity admin. Health page for throttling. Compute page for 14 days of usage.

C10-L4 Sizing a capacity, from trial to the right F SKU

Measure a real peak on a trial, compare it with each SKU's 30-second budget plus headroom, check the F64 licence floor, then reserve only once the load is steady.

Capacity units of an F SKU	The number in its name: F2 has 2, F64 has 64. Trial capacity has 64.
Budget per 30-second window	CUs times 30. An F64 has 1,920 CU-seconds per window.
The F64 licence floor	Below F64, every Power BI viewer needs Pro or PPU. From F64, free-licence users with a viewer role can view.
Sizing method	Trial, read the peak in the Metrics app, pick the smallest SKU covering peak plus headroom, scale and check.